MIL-S-19500/110C
11 April 1972
SUPERSEDING
MIL-S-19500/110B
8 November 1965

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER TYPES 2N328A AND 2N329A

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a PNP, silicon, low-power transistor.

1.2 Physical dimensions. See figure 1 (TO-5).

1.3 Maximum ratings.

P_T 1/	V_{CBO}	V_{EBO}	V_{CEO}	T_{stg} and T_{op}
mW	Vdc	Vdc	Vdc	°C
400	-50	-30	-40 (2N328A) -35 (2N329A)	-65 to +200

1/ Derate linearly 2.28 mW/°C for $T_A > 25°C$.

1.4 Primary electrical characteristics.

Limits	h_{FE}		$V_{CE(sat)}$		$V_{BE(sat)}$		f_{hfb}	
	V_{CE} = -0.5 Vdc I_C = -3 mAdc		I_C = -10 mAdc I_B = -2 mAdc	I_C = -15 mAdc I_B = -2 mAdc	I_C = -3 mAdc I_B = -1 mAdc		V_{CB} = -5 Vdc I_E = 3 mAdc	
	2N328A	2N329A	2N328A	2N329A	2N328A	2N329A	2N328A	2N329A
			Vdc	Vdc	Vdc	Vdc	kHz	kHz
Min	18	36	---	---	---	---	200	250
Max	44	88	-0.3	-0.4	-1	-1	---	---

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specification, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. Transistors shall be of the design, construction, and physical dimensions shown on figure 1.

* 3.3.1 Lead material and finish. Lead material shall be Kovar or alloy 52. Lead finish shall be gold-plated. (Leads may be tin-coated if specified in the contract or order, and this requirement shall not be construed as adversely affecting the qualified-product status of the device, or applicable JAN marking (see 6.2)).

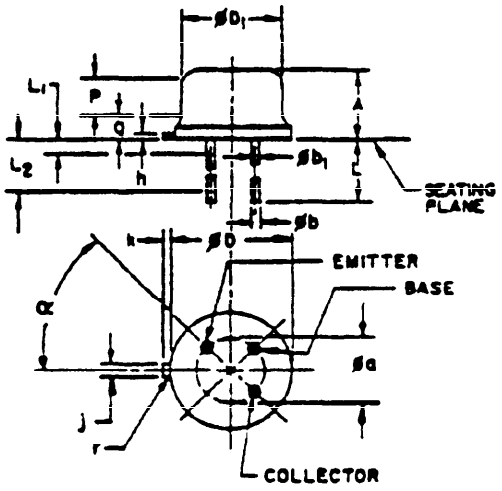
* 3.3.1.1 Selectivity of lead material. Where choice of lead material (see 3.3.1 above) is desired, it shall be specified in the contract or order (see 6.2).

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

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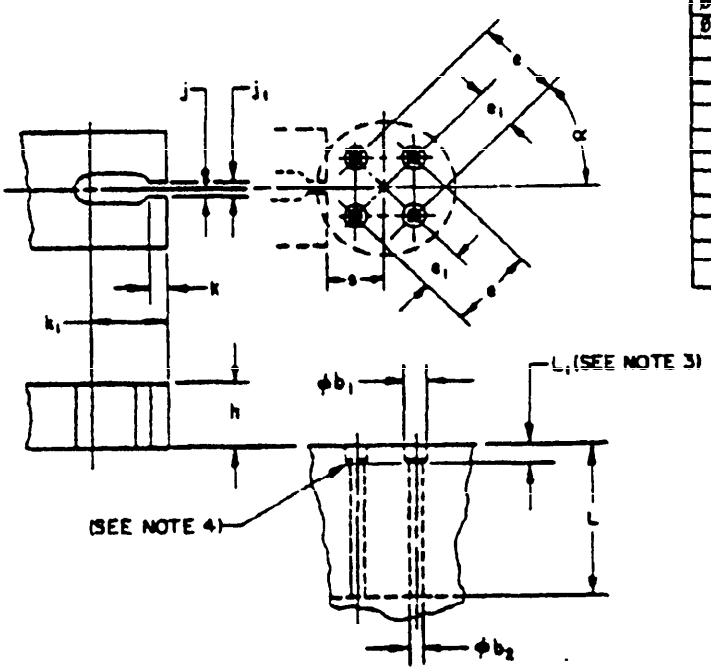


LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.240	.260	6.10	6.60	
øa	.200 TP		5.08 TP		6
øb	.016	.021	.41	.53	7, 8
$øb_1$	.016	.019	.41	.48	7, 8
øD	.335	.370	8.51	9.40	
$øD_1$	.305	.335	7.75	8.51	
h	.009	.041	.23	1.04	
j	.028	.034	.71	.86	2
k	.029	.045	.74	1.14	3
L	1.500	1.750	38.10	44.45	7, 8
L_1	---	.050	---	1.27	7, 8
L_2	.250	---	6.35	---	7, 8
P	.100	---	2.54	---	5
Q	---	.050	---	1.27	4
r	---	.010	---	.25	10
α	45° TP		45° TP		6

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Beyond r(radius) maximum, j shall be held for a minimum length of .011(.28 mm).
3. k measured from maximum øD.
4. Outline in this zone is not controlled.
5. $øD_1$ shall not vary more than .010(.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gage plane .054 + .001-.000(1.37+.03-.00 mm) below seating plane shall be within .007(.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gage and gaging procedure shown in figure 2.
7. $øb_1$ applies between L_1 and L_2. øb applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond L minimum.
8. All three leads.
9. All leads electrically insulated from the case.
10. r(radius) applies to both inside corners of tab.

* FIGURE 1. Physical dimensions of transistor types 2N328A and 2N329A (TO-5).

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
ϕb_1	.0995	.0805	1.51	1.54	
ϕb_2	.0325	.0335	.83	.85	
e	.1995	.2005	5.07	5.09	
e_1	.0995	.1005	2.53	2.55	
h	.150 Nominal		3.81 Nominal		
j	.0175	.0180	.44	.46	-
j1	.0350	.0355	.89	.90	-
k	.009	.011	.23	.28	
k_1	.125 Nominal		3.18 Nominal		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
S	.182	.199	4.62	5.05	1
α	44.90°	45.10°	44.90°	45.10°	

NOTES:
1. The location of the tab locator within the limits indicated will be determined by the tab and flange dimensions of the device being checked.
2. The following gaging procedure shall be used:

   The device being measured shall be inserted until its seating plane is .125(3.18 mm) ±.010(.25 mm) from the seating surface of the gage. A force of 8 ± .5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage.

   The use of a pin straightener prior to insertion in the gage is permissible.

3. Gaging plane.
4. Drill angle.

FIGURE 2. Gage for lead and tab location for transistor types 2N328A and 2N329A.

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TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -50 Vdc		I_{CBO}	---	-0.1	µAdc
Collector to emitter voltage (saturated)	3071	I_B = -2 mAdc		$V_{CE(sat)}$			
2N328A		I_C = -10 mAdc			---	-0.3	Vdc
2N329A		I_C = -15 mAdc			---	-0.4	Vdc
Forward-current transfer ratio	3076	VCE = -0.5 Vdc; I_C = -3 mAdc		h_{FE}			
2N328A					18	44	---
2N329A					36	88	---
Subgroup 3			5				
Emitter to base cutoff current	3061	Bias cond. D; V_{EB} = -30 Vdc		I_{EBO}	---	-.05	µAdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -30 Vdc		I_{CBO}	---	-.03	µAdc
Breakdown voltage, collector to emitter	3011	Bias cond. D; IC = -50 µAdc		BV_{CEO}			
2N328A					-40	---	Vdc
2N329A					-35	---	Vdc
Small-signal short-circuit forward-current transfer-ratio cutoff frequency	3301	V_{CB} = -5 Vdc I_E = 3 mAdc		f_{hfb}			
2N328A					200	---	kHz
2N329A					250	---	kHz
Subgroup 4			5				
Base emitter voltage (saturated)	3066	Test cond. A; I_C = -3 mAdc; I_B = -1 mAdc		$V_{BE(sat)}$	---	-1.0	Vdc
Base emitter voltage (saturated)	3066	Test cond. A; I_C = -10 mAdc; I_B = -2 mAdc		$V_{BE(sat)}$	---	-1.5	Vdc
Forward-current transfer ratio	3076	V_{CE} = -1.5 Vdc; I_B = -1 mAdc		h_{FE}			
2N328A					9	36	---
2N329A					15	72	---
Open circuit output capacitance	3236	V_{CB} = -6 Vdc; I_E = 0; 100 kHz $\leq f \leq$ 1 MHz		C_{obo}	---	110	pF

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TABLE I. Group A inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
• Subgroup 5			10				
High-temperature operation:		T_A = +125°C					
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -30 Vdc		I_{CBO}	---	-10	µAdc
Low-temperature operation:		T_A = -65°C					
Forward-current transfer ratio	3076	V_{CE} = -0.5 Vdc; I_C = -3 mAdc		h_{FE}			
2N328A					9	---	---
2N329A					18	---	---

TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Physical dimensions	2066	See figure 1		---	---	---	---
• Subgroup 2			15				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. C		---	---	---	---
Thermal shock (glass strain)	1056	Test condition A		---	---	---	---
Hermetic seal	1071	Test cond. G or H for fine leaks; test cond. A, C, D, or F for gross leaks		---	---	$1x10^{-7}$	atm cc/s
Moisture resistance	1021	Omit initial conditioning		---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -30 Vdc		I_{CBO}	---	-.03	µAdc
Forward-current transfer ratio	3076	V_{CE} = -0.5 Vdc; I_C = -3 mAdc		h_{FE}			
2N328A					18	44	---
2N329A					36	88	---
• Subgroup 3			15				
Shock	2016	Nonoperating; 1,500 G. for 0.5 ms, 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 3 - Continued							
Vibration, variable frequency	2056	Nonoperating		---	---	---	---
Constant acceleration	2006	20,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test condition E		---	---	---	---
End points:							
Hermetic seal	1071	Test cond. G or H for fine leaks; test cond. A, C, D, or F for gross leaks		---	---	$1x10^{-7}$	atm cc/s
Subgroup 5			20				
Salt atmosphere (corrosion)	1041			---	---	---	---
Subgroup 6			10				
High-temperature life (nonoperating)	1032	T_{stg} = +200°C (see 4.3.4)		---	---	---	---
End points:							
Collector to base cutoff	3036	Bias condition D; V_{CB} = -30 Vdc		I_{CBO}	---	-.06	µAdc
Forward-current transfer ratio	3076	V_{CE} = -0.5 Vdc; I_C = -3 mAdc		h_{FE}			
2N328A					14	52	---
2N329A					28	104	---
Subgroup 7			10				
Steady-state operation life	1027	P_T = 400 mW; V_{CB} = -20 Vdc (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6)							

TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Thermal resistance	3151			θ_{J-A}	---	0.44	°C/mW

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TABLE III. Group C inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
* Subgroup 2			10				
Resistance to solvents	---	MIL-STD-202, Method 215 (see 4.4.1)		---	---	---	---
* Subgroup 3			λ = 10				
High-temperature life (nonoperating)	1031	T_{stg} = +200°C (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							
* Subgroup 4			λ = 10				
Steady-state operation life	1026	P_T = 400 mW; V_{CB} = -20 Vdc; (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S 19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of group A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

* 4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B, 340-hour life-test, may be continued on test to 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C 1,000-hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria, see 4.3.3.

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4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III, and as follows:

* 4.4.1 Resistance to solvents. Transistors shall be subjected to tests in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) All areas of the transistor body where marking has been applied shall be brushed.
(b) After subjection to the tests, there shall be no evidence of mechanical damage to the device and markings shall have remained legible.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

* 6.2 Ordering data.

(a) Lead finish if other than gold-plated (see 3.3.1).
(b) Selectivity of lead material (see 3.3.1.1).

* 6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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